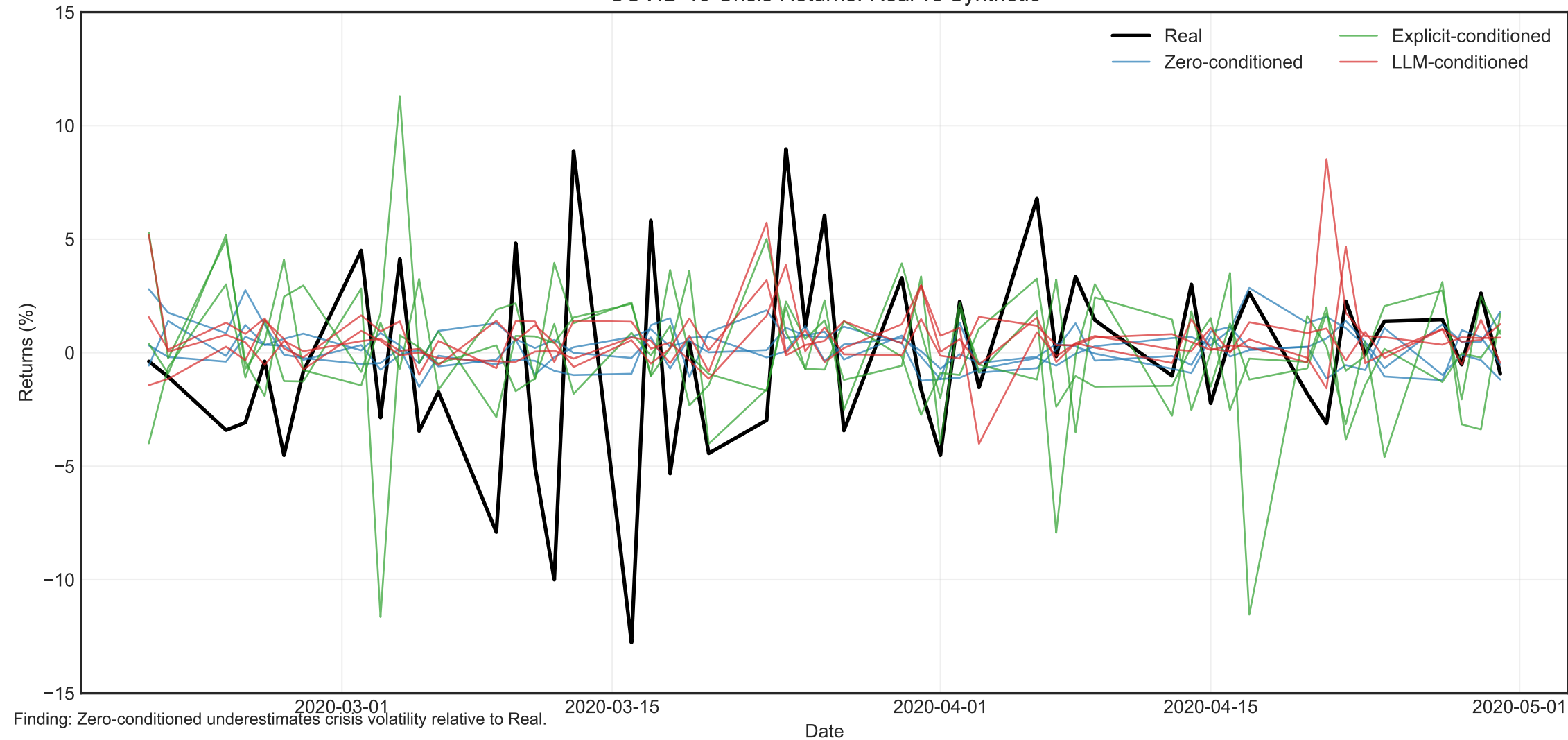
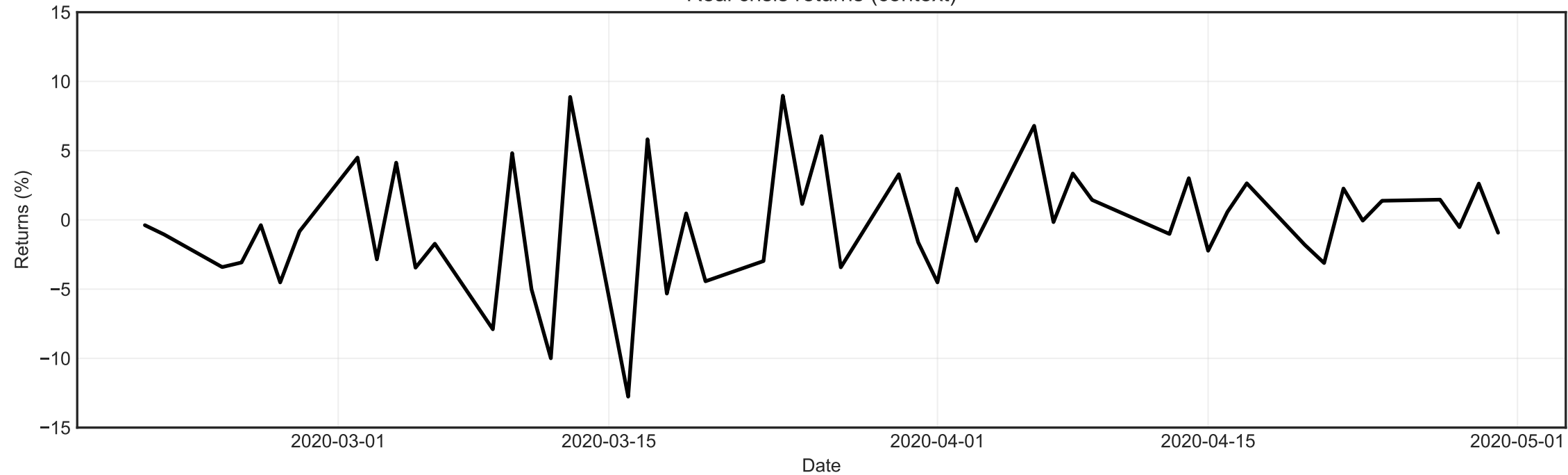


COVID-19 Crisis Returns: Real vs Synthetic

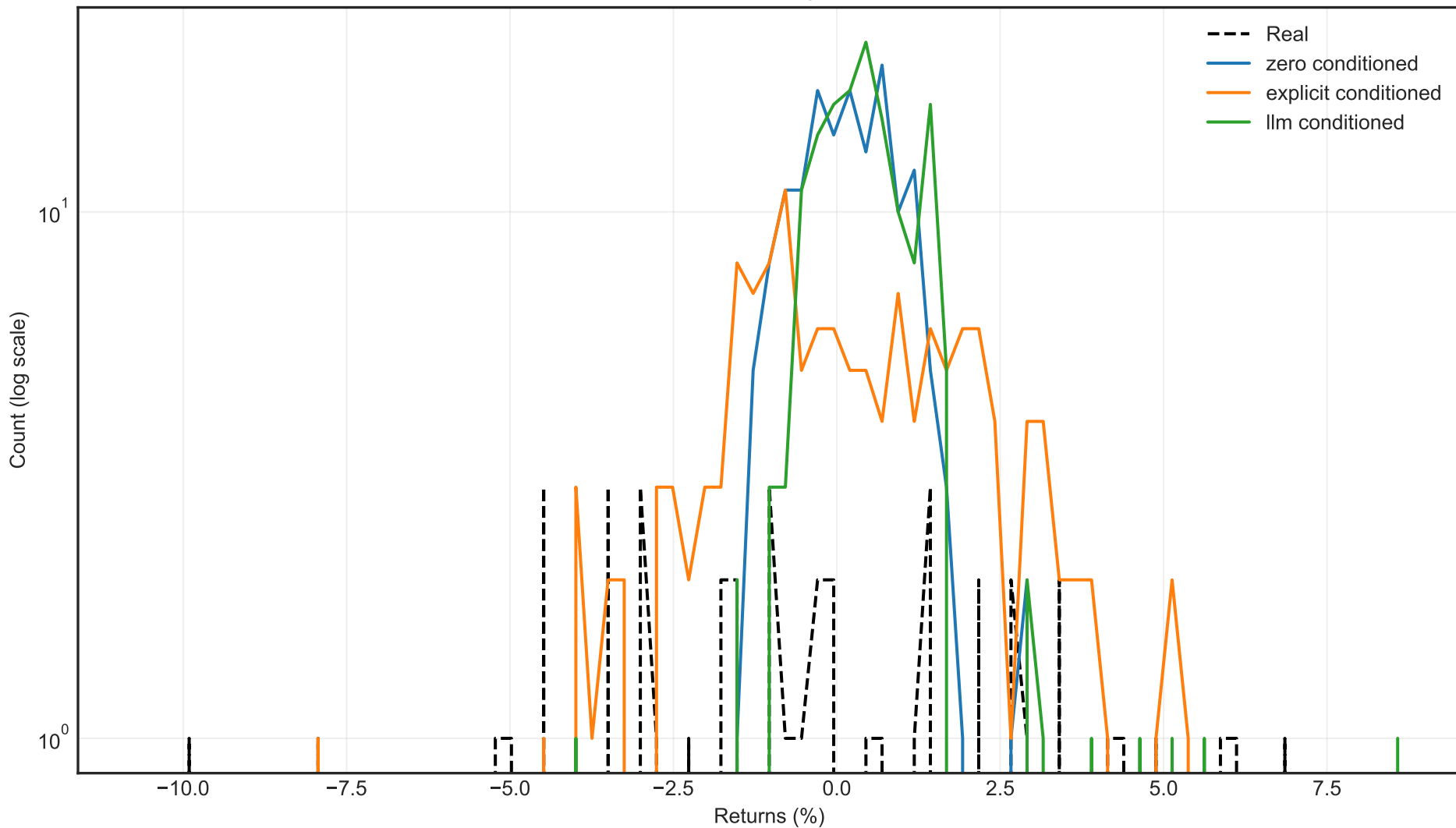


Real crisis returns (context)



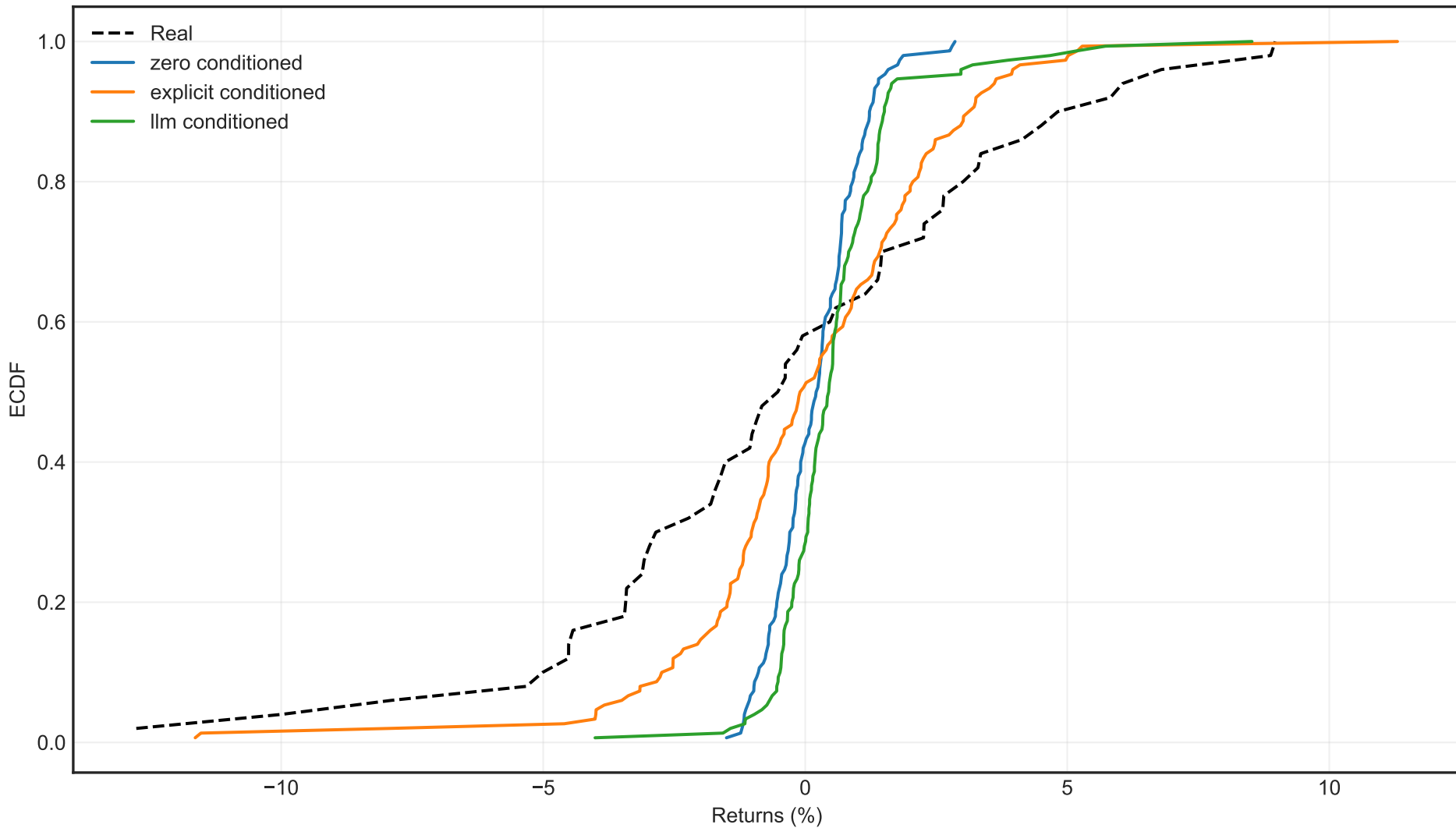
COVID-19 window (2020-02-20 to 2020-04-30); returns in %.

Crisis Histogram



COVID-19 window 2020-02-20 to 2020-04-30; histogram y-axis is log; units are %.

Crisis ECDF Overlay



COVID-19 window 2020-02-20 to 2020-04-30; units are %.

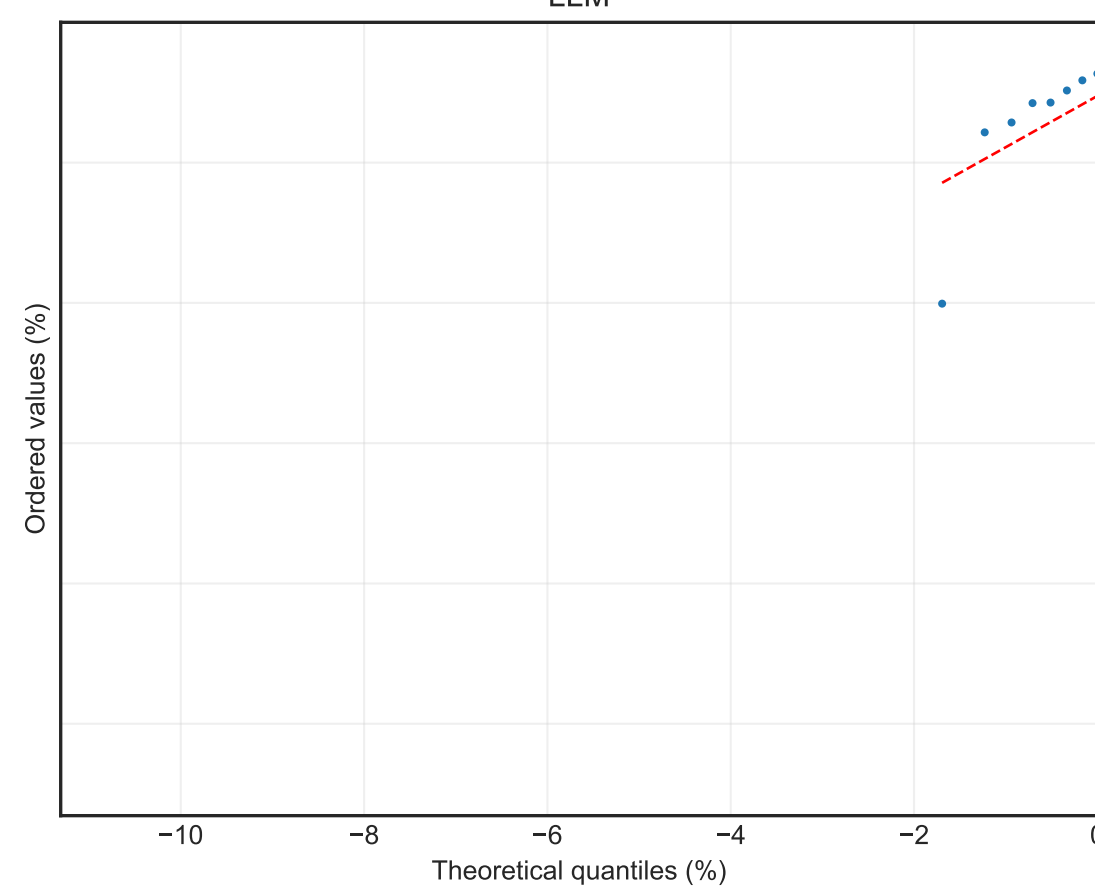
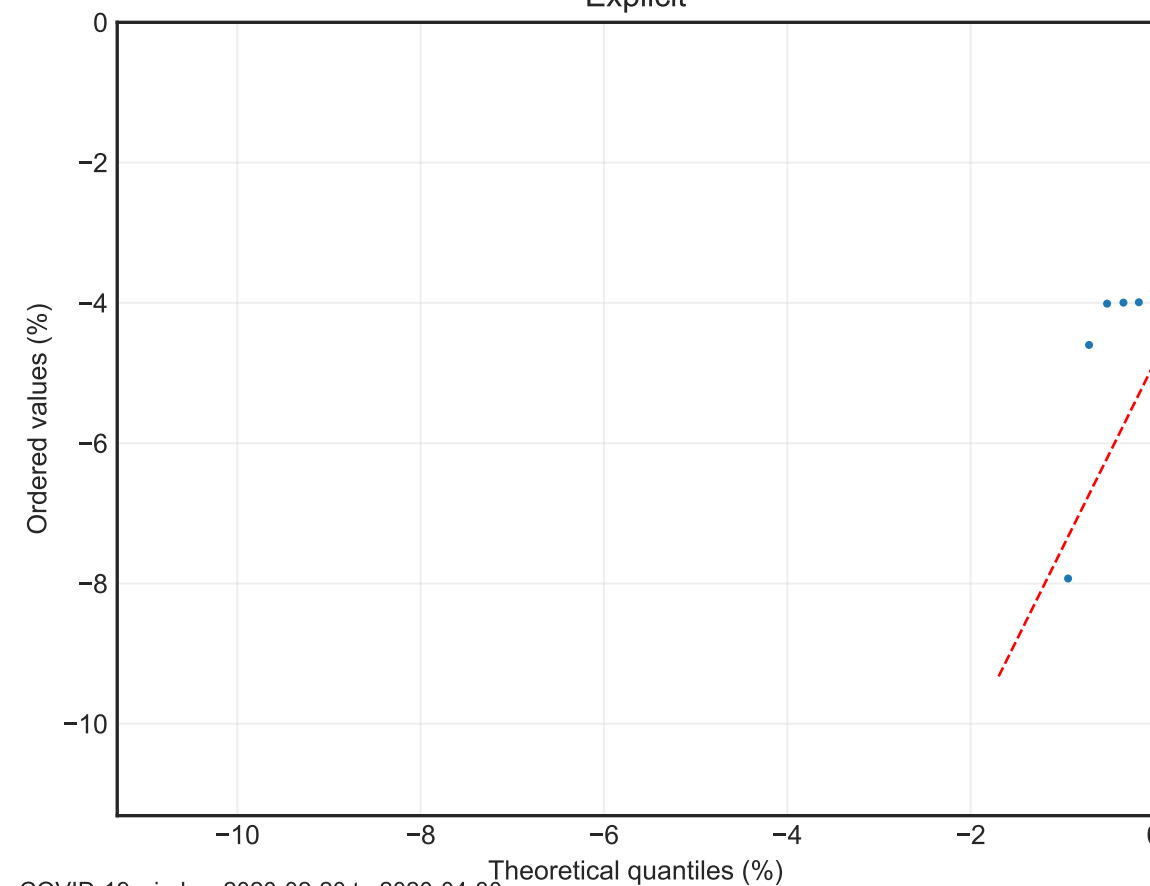
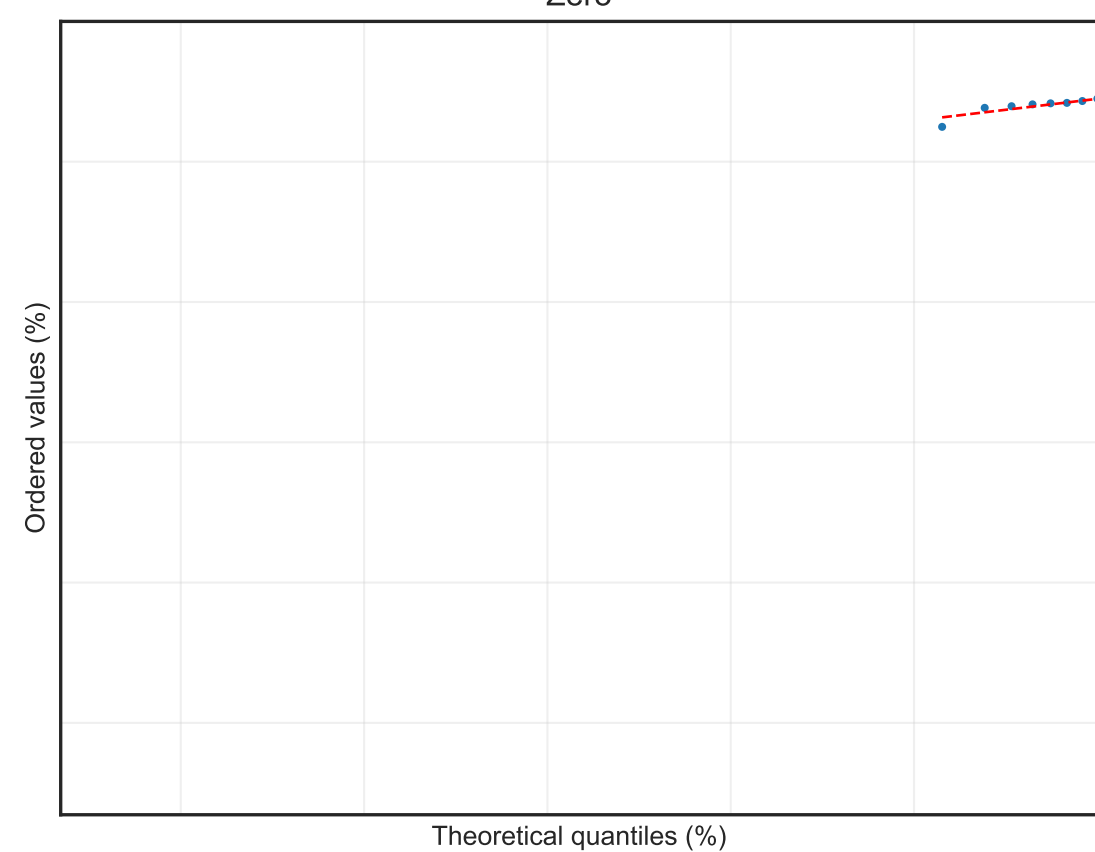
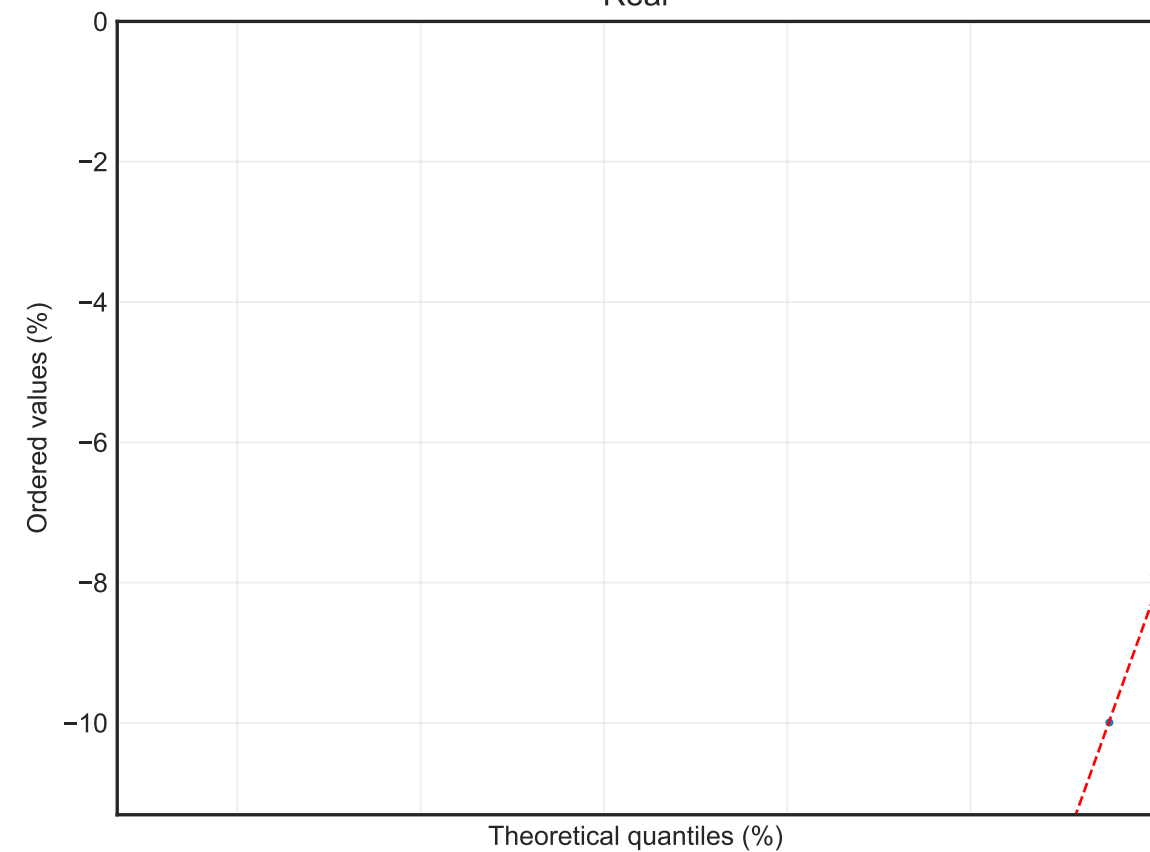
Q-Q Left Tail (shared axes, units: %)

Real

Zero

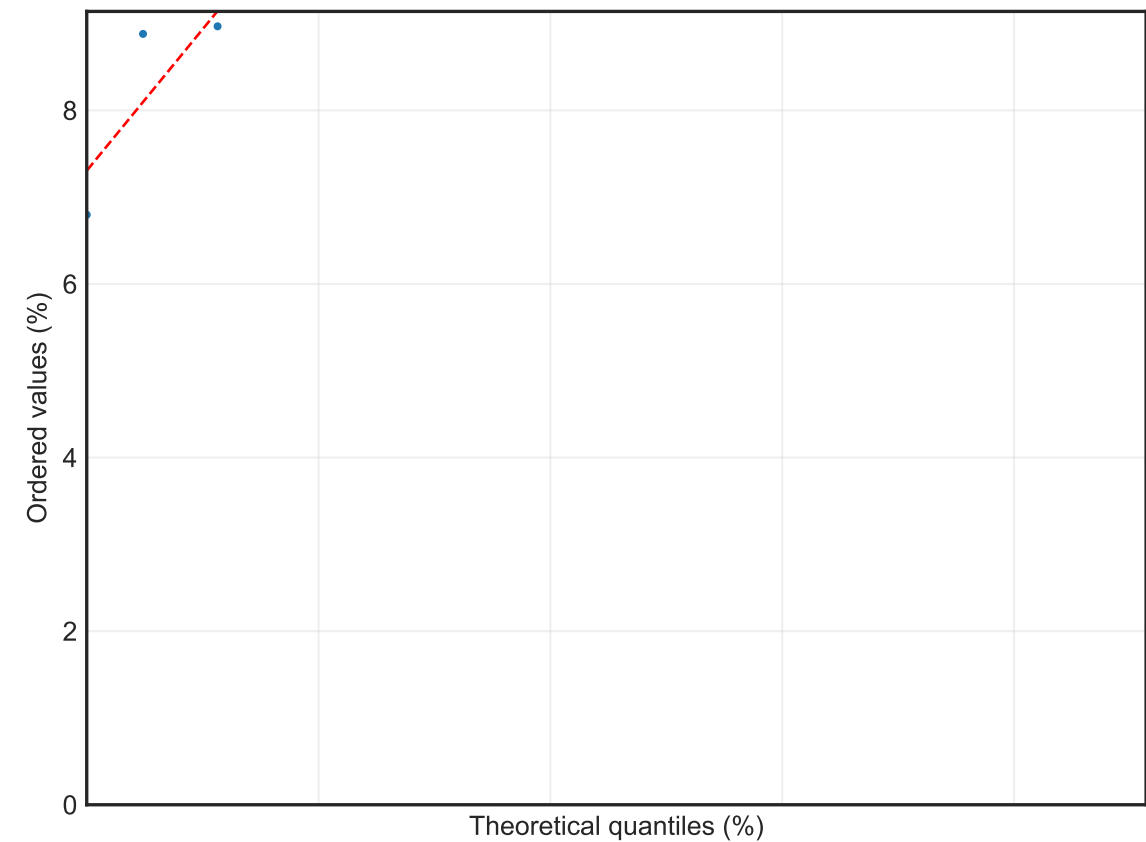
Explicit

LLM

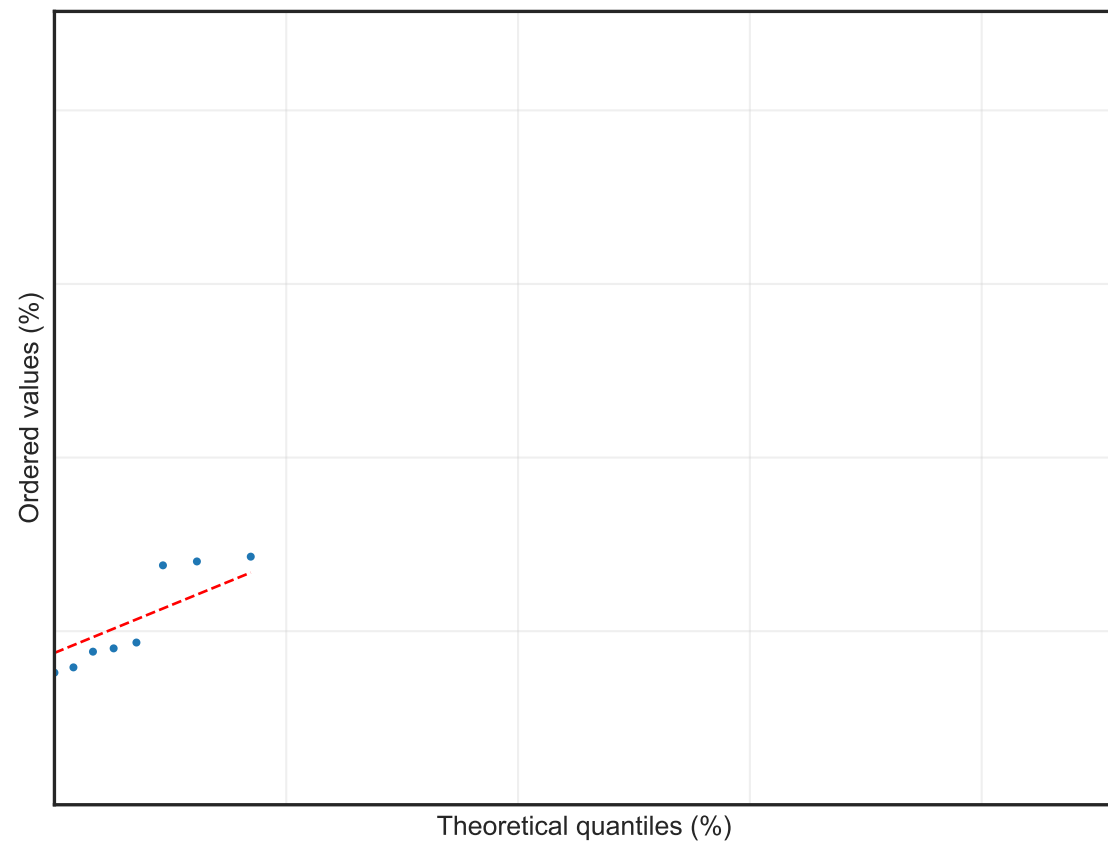


Q-Q Right Tail (shared axes, units: %)

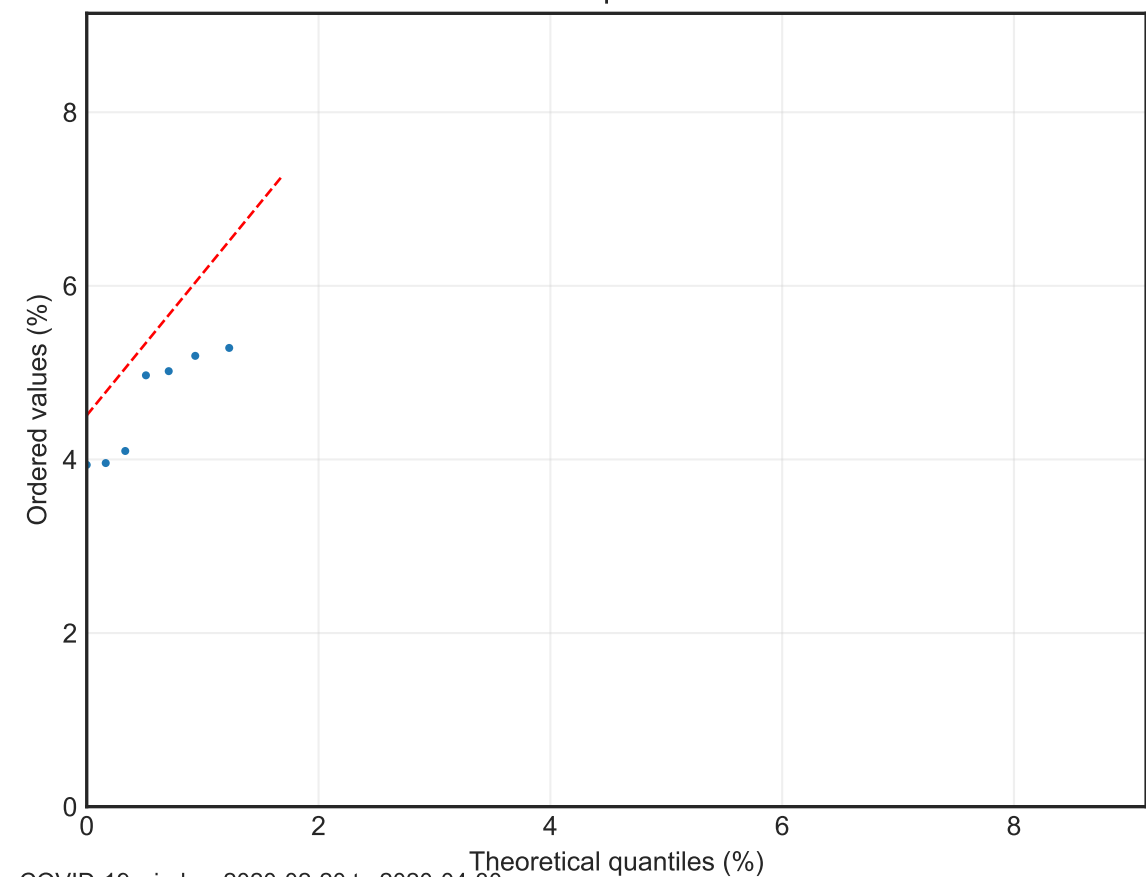
Real



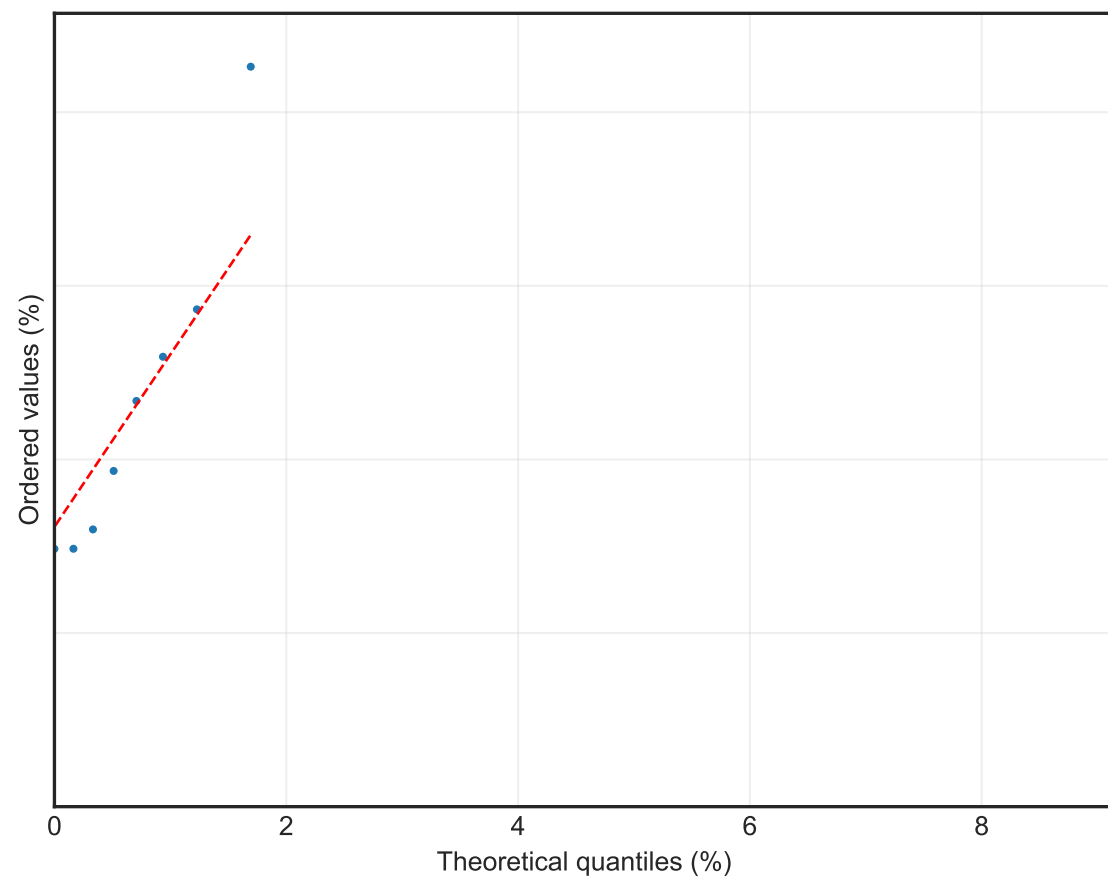
Zero



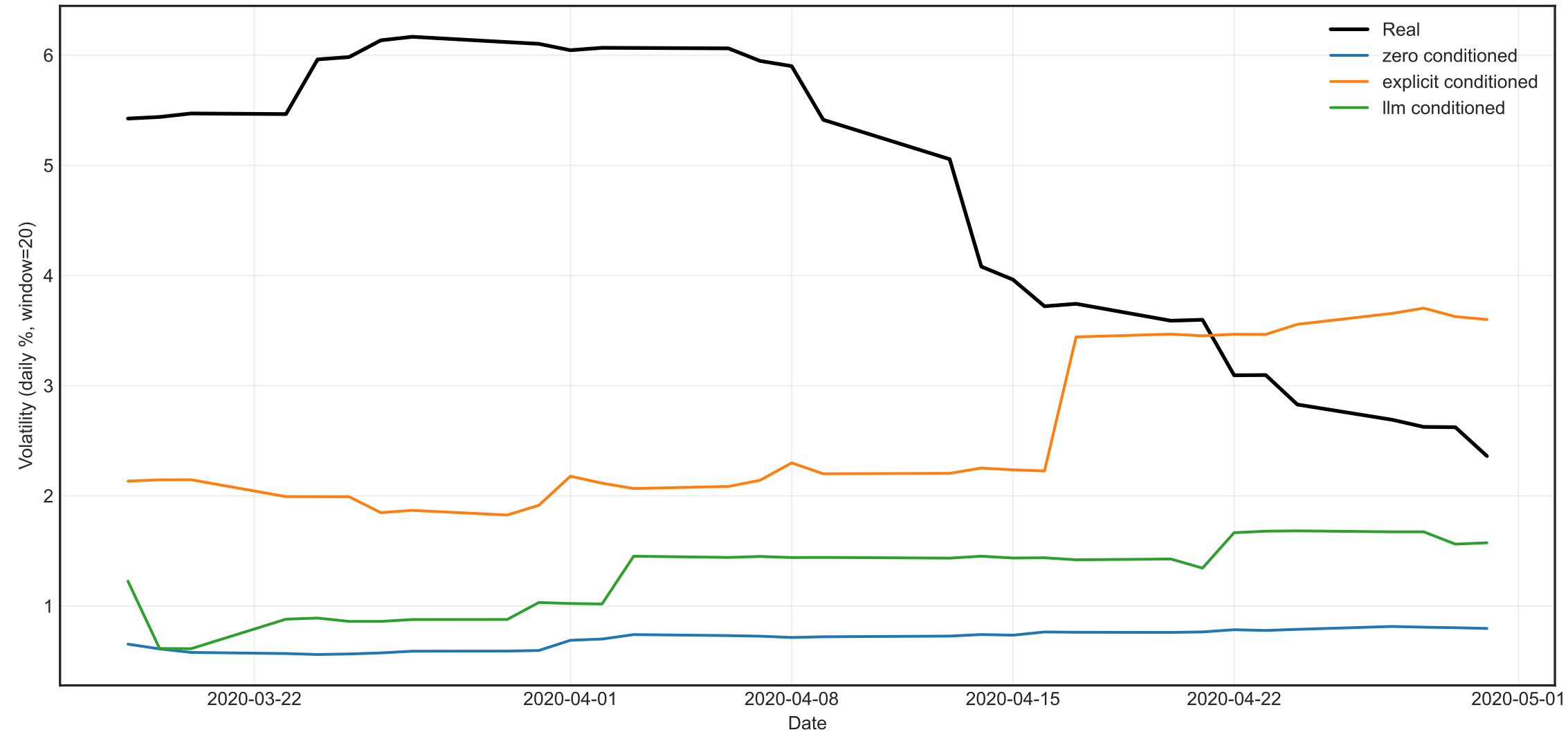
Explicit



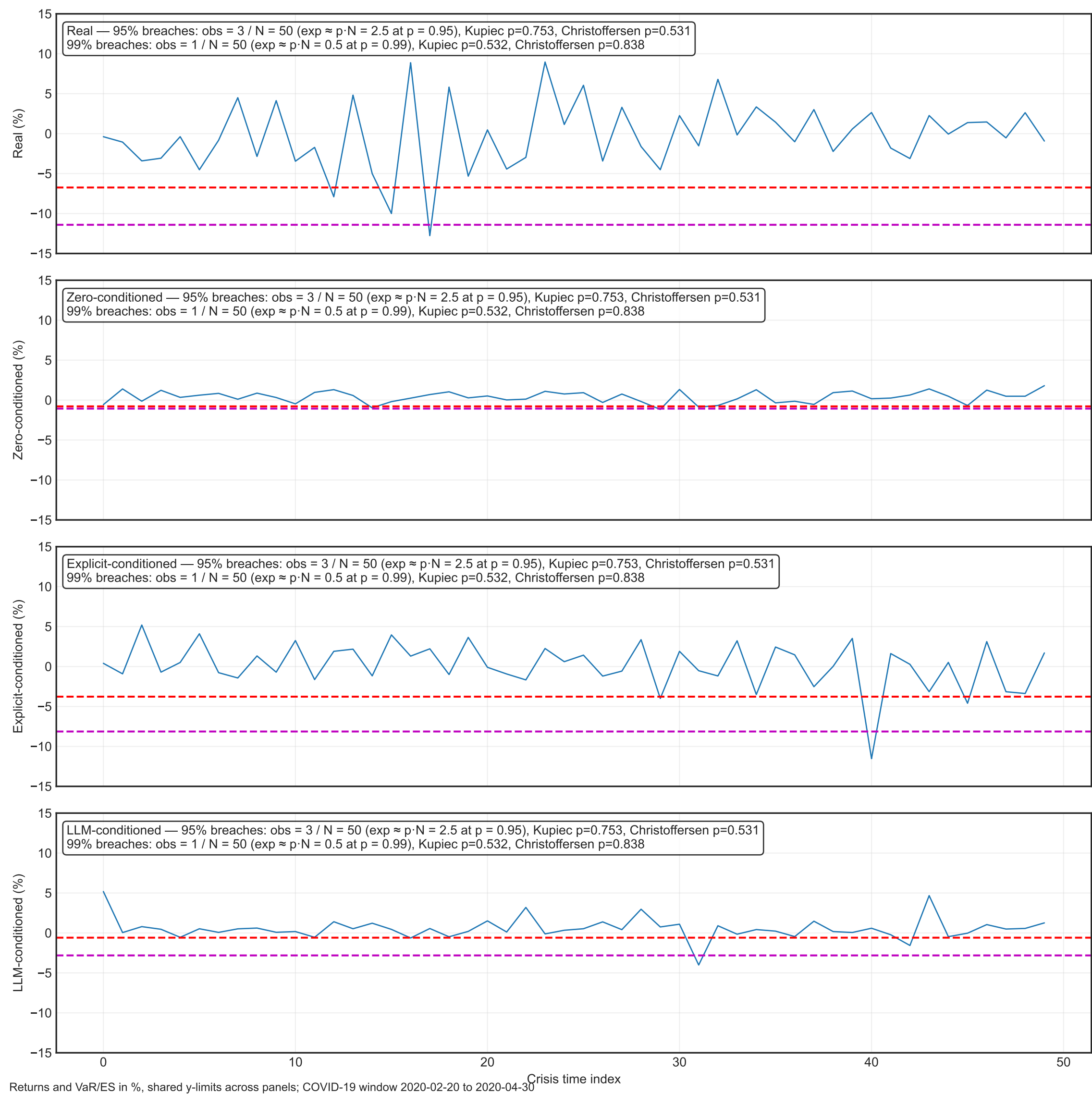
LLM



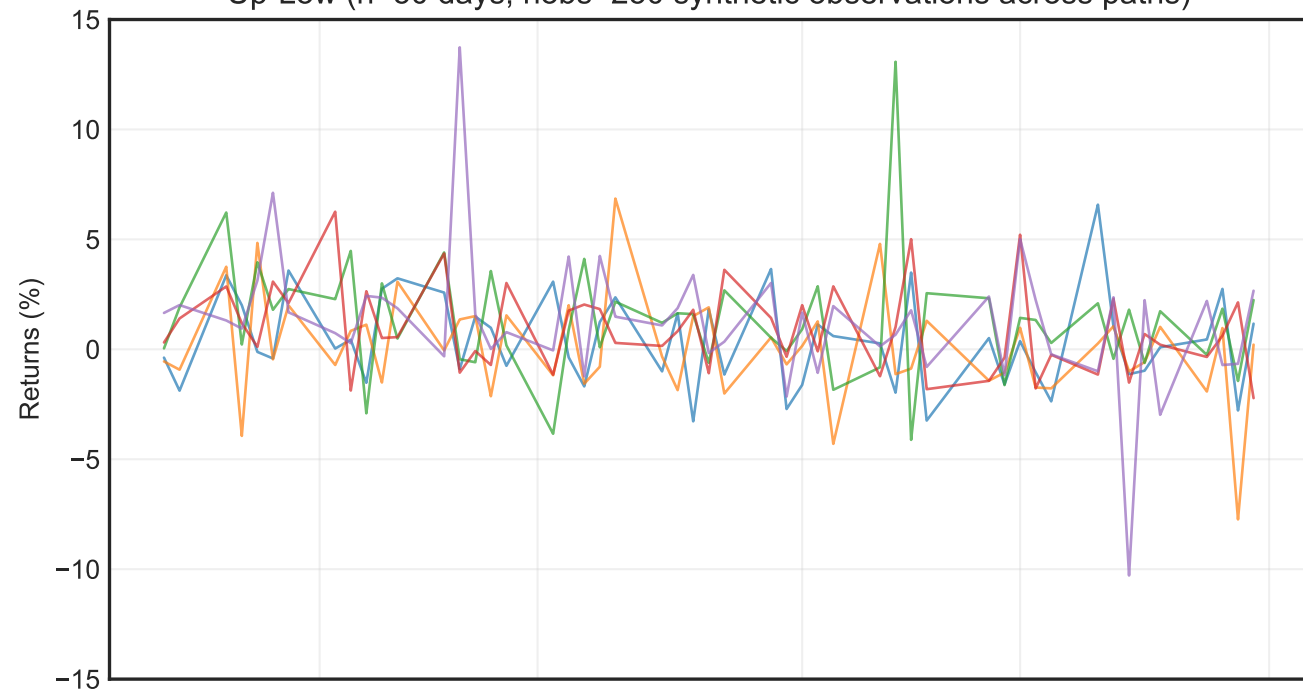
Crisis Rolling Volatility (daily %, window=20, not annualized)



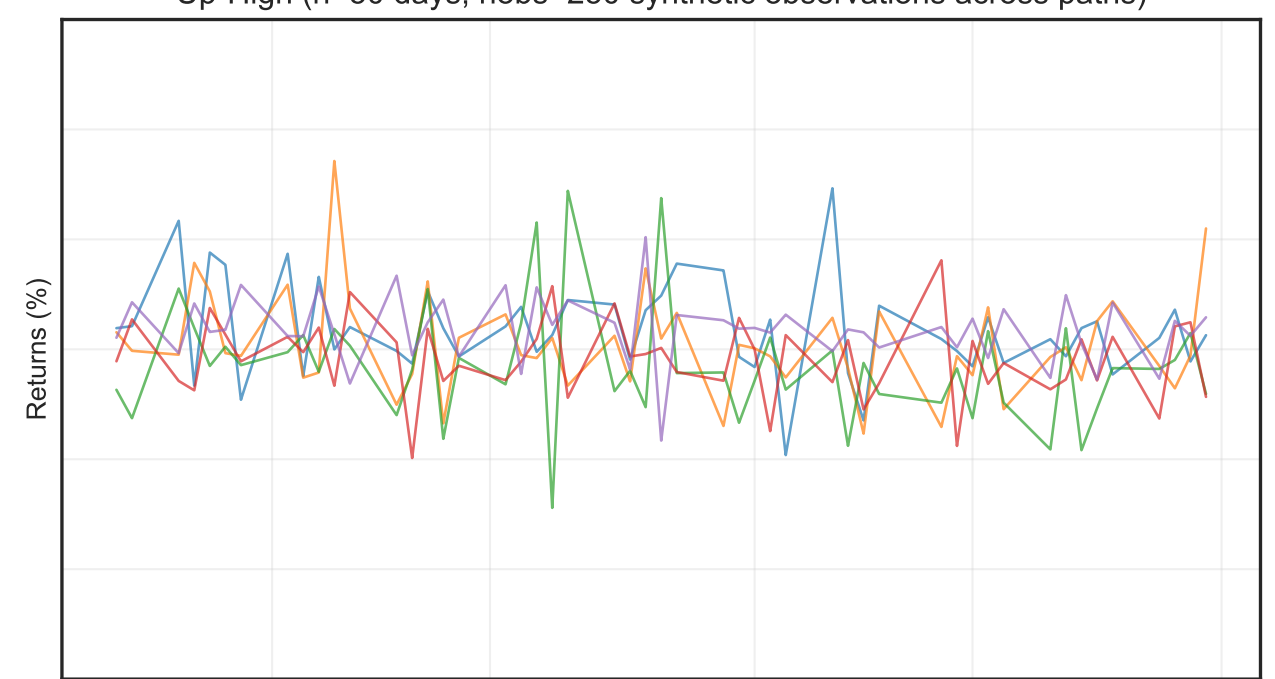
COVID-19 window 2020-02-20 to 2020-04-30; units are daily %; not annualized.



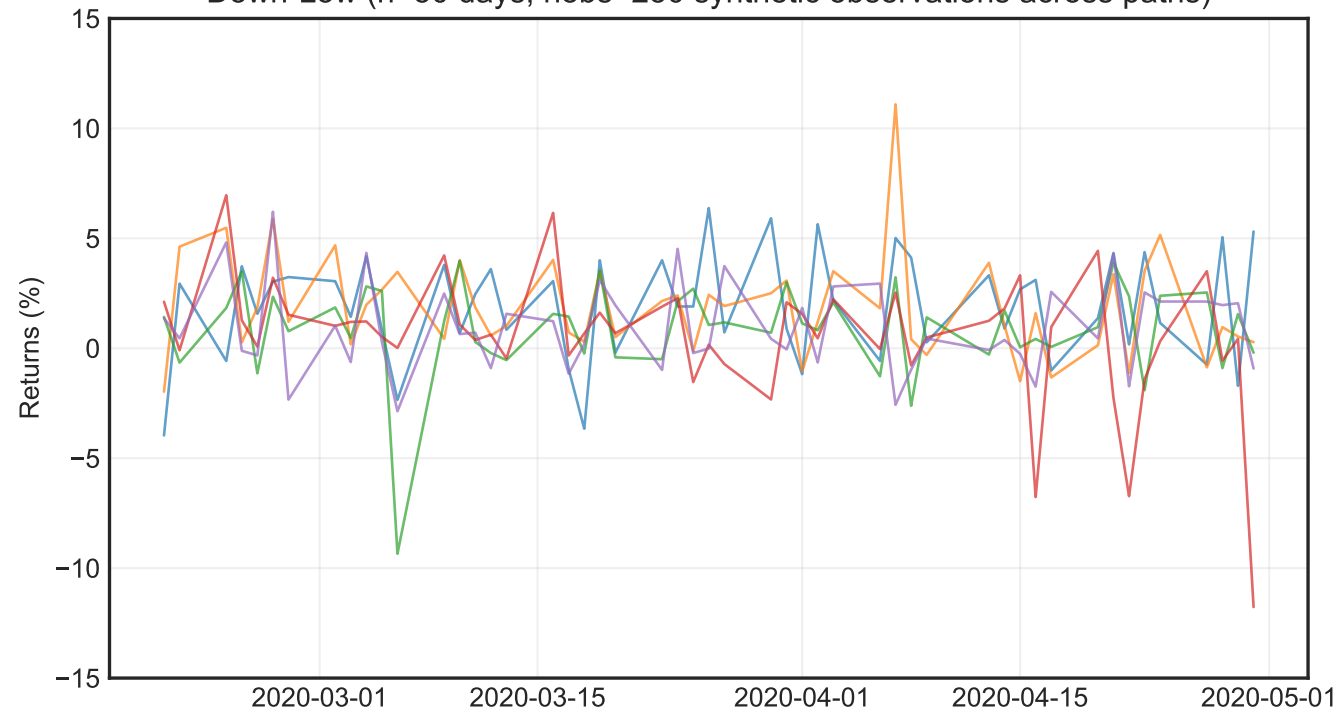
Up-Low (n=50 days, nobs=250 synthetic observations across paths)



Up-High (n=50 days, nobs=250 synthetic observations across paths)



Down-Low (n=50 days, nobs=250 synthetic observations across paths)



Down-High (n=50 days, nobs=250 synthetic observations across paths)

